JNTUK R23 · SEMESTER I

Basic Electrical & Electronics Engineering

Unit 1: DC & AC Circuits

Subject: BEEE (Basic Electrical & Electronics Engineering)

Regulation: JNTUK R23 | **Semester:** I (First Year)

Branch: Common to All Engineering Branches

Reference: D.C. Kulshreshtha (TMH) · Rajendra Prasad (PHI) · D.P. Kothari & I.J. Nagrath

- ⚡ Electrical Circuit Elements (R, L, C)
- 📐 Ohm's Law & Kirchhoff's Laws
- 🔗 Series, Parallel & Series-Parallel Circuits
- 🔄 Superposition Theorem
- ∼ AC Fundamentals & Waveforms
- 📊 RMS, Average, Form & Peak Factors
- 📡 Phasor Diagrams (R, L, C)
- 🔋 Active, Reactive & Apparent Power

1 Course Objectives

By the end of this unit, students will be able to:

- ▶ Understand and differentiate the three passive circuit elements — Resistor (R), Inductor (L), and Capacitor (C) — and their energy behaviour in DC and AC circuits.
- ▶ Apply Ohm's Law and Kirchhoff's Laws (KCL & KVL) to analyse simple and complex DC networks through systematic methods.
- ▶ Solve series, parallel, and series-parallel resistive circuits to find equivalent resistance, branch currents, and voltage drops.
- ▶ Apply the Superposition Theorem to circuits containing multiple independent sources and compute responses efficiently.
- ▶ Analyse AC sinusoidal waveforms and compute key parameters — time period, frequency, amplitude, phase, average value, RMS value, form factor, and peak factor.
- ▶ Draw and interpret phasor diagrams for pure R, L, and C circuits; understand the concept of impedance, reactance, and power factor in AC circuits.
- ▶ Distinguish between Active Power (W), Reactive Power (VAR), and Apparent Power (VA) and apply the power triangle concept to practical AC circuits.

2 Mind Map – Unit 1 Overview

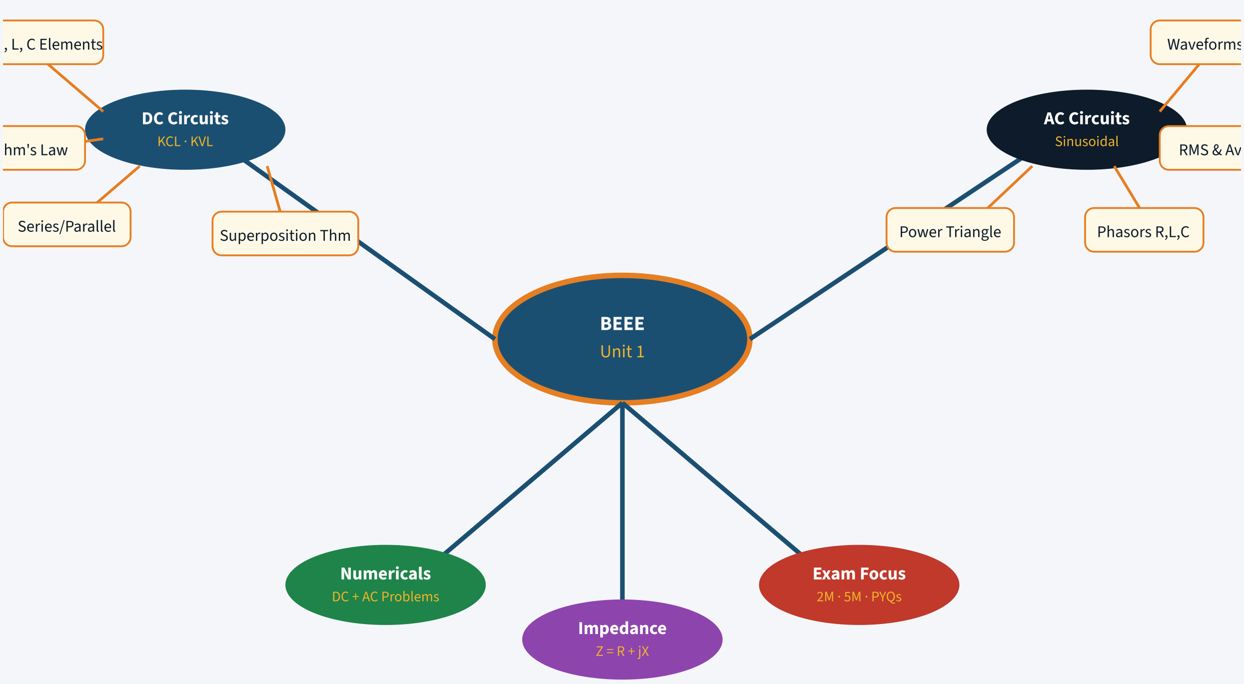


Fig: Mind Map – BEEE Unit 1 Topic Connections

3 Required Background Knowledge

⚡ Charge & Current

Electric charge (Q) is the fundamental property of matter. Current (I) is the rate of flow of charge.

```
I = dQ/dt    [Unit: Ampere (A)]
```

🔋 Voltage & EMF

Voltage (V) is the electric potential difference between two points. It drives current in a circuit.

```
V = W/Q    [Unit: Volt (V)]
```

📐 Basic Ohm's Law

$V = IR$: Voltage is proportional to current for a fixed resistance at constant temperature. R is in Ohms (Ω).

📊 Trigonometry for AC

AC quantities use sin/cos functions: $\sin^2\theta + \cos^2\theta = 1$, $\sin(\omega t \pm \pi/2) = \pm\cos(\omega t)$. Essential for phasor analysis.

4 Detailed Theory – DC Circuits

4.1 Electrical Circuit Elements

A. Resistor (R)

A resistor creates electrical friction to **limit or control the flow of electric current** in a circuit.

- **Function:** Limits current flow; opposes it.
- **Energy Handling:** Dissipates electrical energy as **heat** (irreversibly).
- **SI Unit:** Ohm (Ω)
- **Symbol:** Zig-zag line (—/\/\/—)

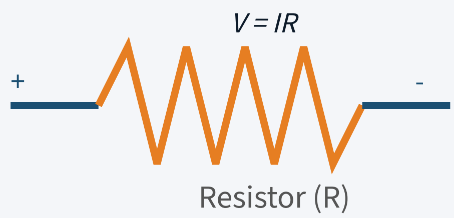


Fig: Resistor symbol and circuit

B. Capacitor (C)

A capacitor **stores electrical energy in an electric field**. It consists of two parallel plates separated by a dielectric (insulating) material.

- **Function:** Stores charge; blocks DC once fully charged, allows AC.
- **Energy Handling:** Stores and releases energy in electric field.
- **SI Unit:** Farad (F)
- **Symbol:** Two parallel lines (—| |—)
- **Relation:** $Q = CV$, where C = capacitance

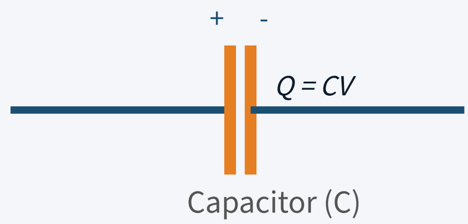


Fig: Capacitor symbol and circuit

C. Inductor (L)

An inductor **stores energy in a magnetic field**. It is typically a coil of wire.

- **Function:** Stores energy in magnetic field; opposes change in current.
- **Energy Handling:** Offers little opposition to steady DC; strongly opposes AC.
- **SI Unit:** Henry (H)
- **Symbol:** Series of coils or loops
- **Relation:** $V = L(di/dt)$

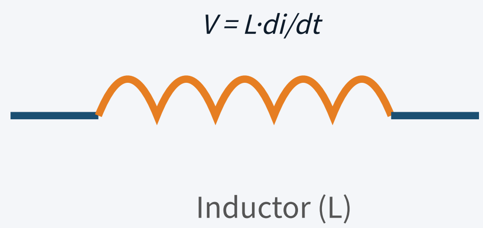


Fig: Inductor symbol and circuit

Comparison of R, L, C Elements

Property	Resistor (R)	Inductor (L)	Capacitor (C)
Energy Storage	No (dissipates)	Magnetic field	Electric field
V-I Relation	V = IR	V = L(di/dt)	I = C(dv/dt)
DC Behaviour	Conducts	Short circuit (wire)	Open circuit
AC Behaviour	Constant R	Opposes (X_L=ωL)	Opposes (X_C=1/ωC)
Unit	Ohm (Ω)	Henry (H)	Farad (F)
Phase (AC)	V and I in phase	V leads I by 90°	I leads V by 90°

4.2 Ohm's Law

📜 STATEMENT

The current flowing through a conductor is directly proportional to the potential difference across its ends, provided the physical conditions (temperature, pressure) remain constant.

$$V = I \times R \quad | \quad I = V/R \quad | \quad R = V/I$$

Where: V = Voltage (Volts), I = Current (Amperes), R = Resistance (Ohms, Ω)

Limitations of Ohm's Law

- ▶ Applicable only for **good (metallic) conductors**.
- ▶ Valid only when physical conditions like **temperature, pressure, and tension remain constant**.
- ▶ **Not applicable** for semiconductors, thermistors, vacuum tubes, diodes.
- ▶ Does not apply to **non-linear devices** (V-I graph is not a straight line).
- ▶ Not valid for **electrolytes** where chemical changes occur.

💡 **Remember:** Ohm's Law is only valid for linear bilateral passive resistors. Devices that violate it are called "non-ohmic" devices.

4.3 Kirchhoff's Laws

A. Kirchhoff's Current Law (KCL)

📜 STATEMENT

The algebraic sum of all currents meeting at a junction (node) in a circuit is zero.

Sign Convention: Current entering → Positive (+), Current leaving → Negative (−)

$$\Sigma I_{in} = \Sigma I_{out} \quad \text{OR} \quad \Sigma I = 0 \text{ at any node}$$

If six conductors carry currents $I_1, I_2, I_3, I_4, I_5, I_6$ meeting at point O:

$$I_1 + I_6 + I_4 - I_5 - I_3 - I_2 = 0$$

Currents towards O are positive; currents away from O are negative.

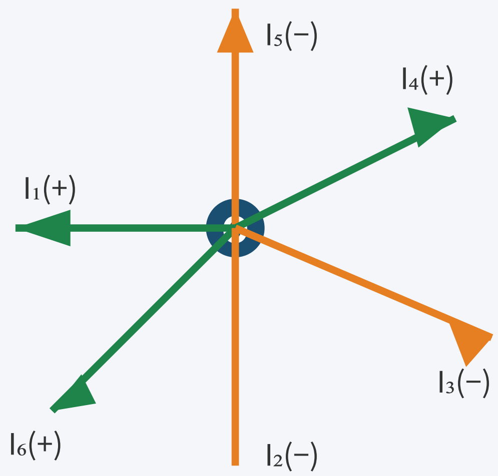


Fig: KCL at node O

B. Kirchhoff's Voltage Law (KVL)

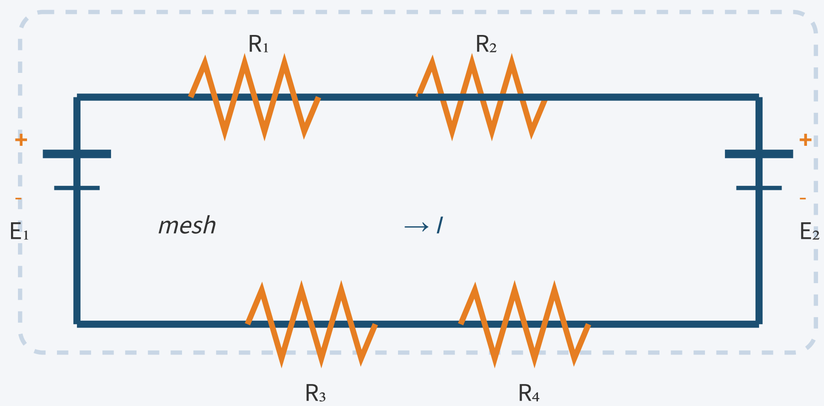
📜 **STATEMENT**

In any closed loop/mesh, the algebraic sum of all EMFs is equal to the algebraic sum of voltage drops across all circuit elements.

$$\Sigma E = \Sigma IR \quad \text{(in any closed loop)}$$

Sign Convention for KVL:

- **EMF:** Rise in potential (+E), Fall in potential (−E)
- **Voltage Drop:** In direction of current → Fall (−IR), Against current → Rise (+IR)



KVL: $E_1 - E_2 = IR_1 + IR_2 + IR_3 + IR_4$

Fig: KVL applied to a closed mesh with two EMF sources

4.4 Series, Parallel & Series-Parallel Circuits

A. Series Circuit

In a series circuit, resistances are connected **end to end** so there is only one path for current to flow.

- **Current:** Same through all elements — $I = I_1 = I_2 = I_3$
- **Voltage:** Divides — $V = V_1 + V_2 + V_3$
- **Equivalent Resistance:** $R_{eq} = R_1 + R_2 + R_3 + ... + R_n$

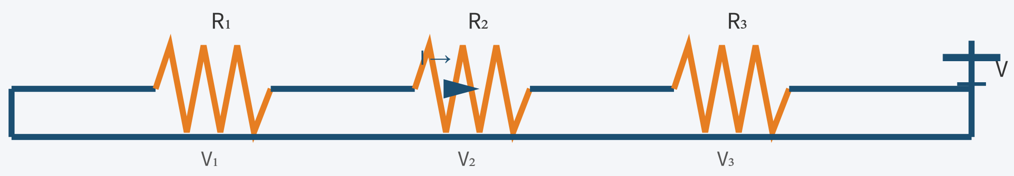


Fig: Series Circuit — $R_{eq} = R_1 + R_2 + R_3$

B. Parallel Circuit

In a parallel circuit, one end of each resistance is joined to a common point (A) and the other end to another common point (B). There are as many paths for current as there are resistances.

- **Voltage:** Same across all branches — $V = V_1 = V_2 = V_3$
- **Current:** Divides — $I = I_1 + I_2 + I_3$
- **Equivalent Resistance:** $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$

C. Series-Parallel Circuit

A series-parallel combination has some components in series (same current) and others in parallel (same voltage). It allows complex networks to achieve desired electrical behaviour.

Parameter	Series Circuit	Parallel Circuit
Current	Same everywhere	Divides among branches
Voltage	Divides	Same across branches
R_{eq}	$R_1+R_2+...+R_n$ (increases)	Less than smallest R
Failure effect	All fail if one fails	Others work if one fails
Applications	Christmas lights, fuses	Home appliances, wiring

4.5 Superposition Theorem

📜 **STATEMENT**

In a linear circuit with multiple independent sources, the voltage/current across any element is the algebraic sum of the voltages/currents produced by each independent source acting alone, with all other sources replaced by their internal resistances.

Procedure (Step-by-Step):

Step 1: Consider one independent source at a time. Replace all *voltage sources* by short circuits (R=0) and *current sources* by open circuits (R=∞). Do NOT turn off dependent sources.

Step 2: Calculate the response (voltage/current) in the desired element due to this single source.

Step 3: Repeat Steps 1 & 2 for each independent source separately.

Step 4: Algebraically add all individual responses. The total is the actual response with all sources active.

⚠ **Limitations:** Superposition Theorem is NOT applicable to (i) power calculations, (ii) non-linear circuits, (iii) circuits with dependent sources (do not deactivate them).

5 Detailed Theory – AC Circuits

5.1 AC Fundamentals

An alternating current (AC) is one that periodically reverses its direction. It can be generated by rotating a coil at constant angular velocity in a uniform magnetic field.

Equation of AC Voltage & Current

$$v(t) = V_m \sin(\omega t) = V_m \sin(2\pi ft)$$

$$i(t) = I_m \sin(\omega t) = I_m \sin(2\pi ft)$$

Where: V_m / I_m = Peak (maximum) value, $\omega = 2\pi f$ = angular frequency (rad/s), f = frequency (Hz), T = time period (s)

5.2 Waveform Parameters

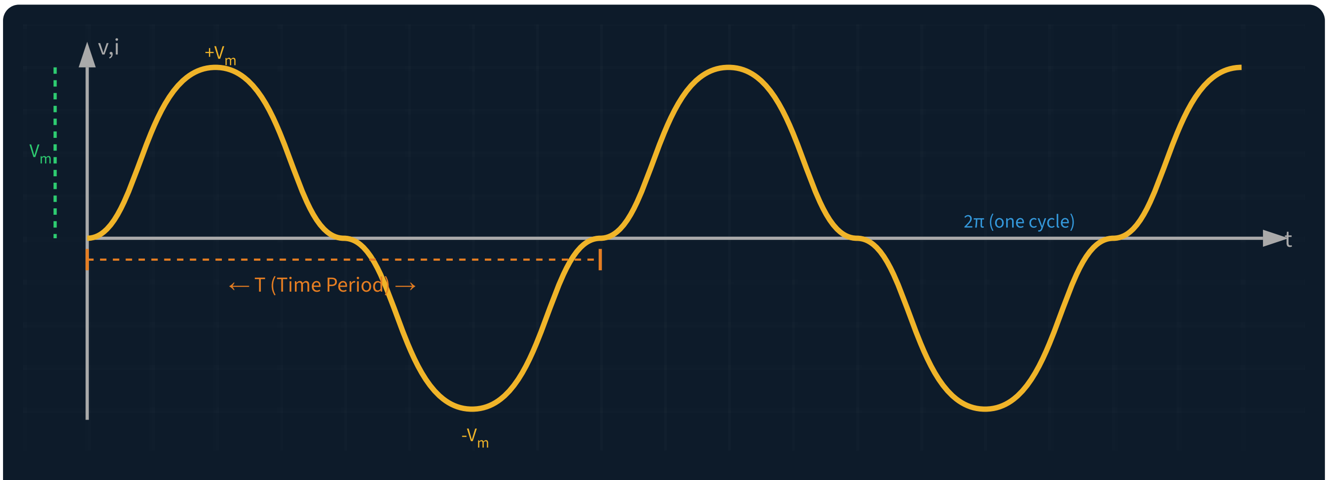


Fig: Sinusoidal AC Waveform — $v(t) = V_m \sin(\omega t)$

Key AC Waveform Parameters

Parameter	Symbol	Definition	Formula/Value
Time Period	T	Time for one complete cycle	T = 1/f seconds
Frequency	f	Cycles per second	f = 1/T Hz
Angular Frequency	ω	Radians per second	$\omega = 2\pi f$ rad/s
Amplitude (Peak)	V_m or I_m	Maximum value (positive half)	$V_m = \sqrt{2} \times V_{rms}$
Peak-to-Peak	V_{pp}	Positive peak to negative peak	$V_{pp} = 2V_m$
Instantaneous Value	v, i	Value at any particular instant	$v = V_m \sin(\omega t)$

Average Value	V_{av}	Average over half cycle (full cycle = 0)	$V_{av} = 2V_m/\pi = 0.637\ V_m$
RMS Value	V_{rms}	Equivalent DC heating value	$V_{rms} = V_m/\sqrt{2} = 0.707\ V_m$

5.3 Average Value – Derivation

The average value of a full cycle AC is **zero** (positive and negative halves cancel). The average is computed over a **half cycle**:

$$I_{avg} = (2/T) \int_0^{T/2} I_m \sin(\omega t)\, dt$$

$$I_{avg} = 2I_m/\pi = 0.637\ I_m$$

5.4 RMS Value – Derivation

The RMS (Root Mean Square) value is the equivalent DC value that produces the same heating effect in a resistor as the AC waveform.

$$I_{rms} = \left[1/T \int_0^T I^2 dt\right]^{1/2}$$

$$I_{rms} = I_m/\sqrt{2} = 0.707\ I_m$$

Derivation Outline: Substituting $I = I_m \sin(\omega t)$ and using $\sin^2(\theta) = (1-\cos 2\theta)/2$, integrating over one full period gives $I_{rms} = I_m/\sqrt{2}$.

5.5 Form Factor & Peak Factor

FORM FACTOR (K_F)

$$K_f = V_{rms} / V_{av} = (V_m/\sqrt{2}) / (2V_m/\pi) = 1.11$$

Ratio of RMS to average value. For sinusoidal wave: **1.11**

PEAK FACTOR (K_P)

$$K_p = V_m / V_{rms} = V_m / (V_m/\sqrt{2}) = \sqrt{2} = 1.414$$

Ratio of peak to RMS value. For sinusoidal wave: **√2 ≈ 1.414**

5.6 Phase & Phase Difference

Phase of an AC quantity is the fractional part of the time period or cycle through which the quantity has advanced from its selected zero position.

Phase Difference: When two alternating quantities of the same frequency have different zero crossing points, they are said to have a phase difference (ϕ).

$$v_1 = V_m \sin(\omega t) \qquad v_2 = V_m \sin(\omega t - \phi) \rightarrow v_2 \text{ lags } v_1 \text{ by } \phi$$

Phasor: A rotating line of definite length rotating anticlockwise at constant angular velocity ω, representing a sinusoidal AC quantity. Length = peak value; angle = phase angle.

5.7 Phasor Diagrams for R, L, C

A. Pure Resistor (R) Circuit

Applying KVL: $v(t) = R \cdot i(t)$

If $i(t) = I_m \sin(\omega t)$, then $v(t) = V_m \sin(\omega t)$

- ▶ Voltage and Current are **in phase** ($\phi = 0°$)
- ▶ Power factor = cos(0°) = **1 (Unity)**
- ▶ All power is consumed (no reactive component)

$$V_m = R \cdot I_m$$

Phasor: Pure R Circuit

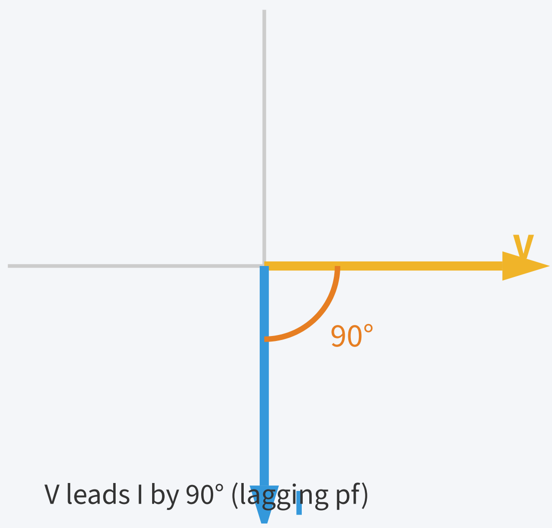
B. Pure Inductor (L) Circuit

EMF across inductor: $e = -L(di/dt)$

If $v(t) = V_m \sin(\omega t)$, then $i(t) = I_m \sin(\omega t - \pi/2)$

- ▶ **Voltage LEADS Current by 90°** (or Current lags Voltage)
- ▶ Inductive Reactance: $X_L = \omega L = 2\pi f L$ (Ω)
- ▶ Power factor = cos(90°) = **0 (lagging)**

$$X_L = \omega L \quad | \quad I_m = V_m / X_L$$



Phasor: Pure L Circuit

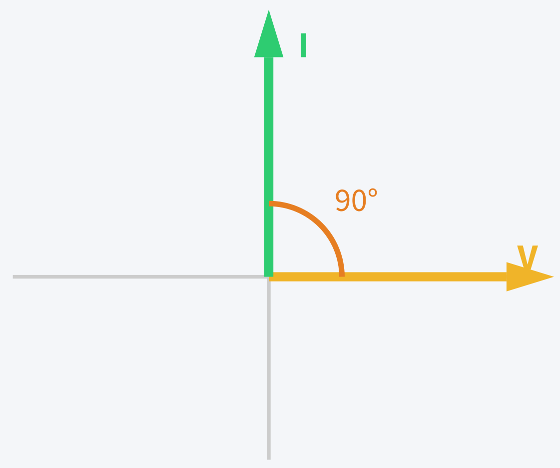
C. Pure Capacitor (C) Circuit

$q = CV_m \sin(\omega t)$, $i(t) = dq/dt = \omega CV_m \cos(\omega t)$

Therefore $i(t) = I_m \sin(\omega t + \pi/2)$

- **Current LEADS Voltage by 90°**
- Capacitive Reactance: $X_C = 1/(\omega C) = 1/(2\pi fC)$ (Ω)
- Power factor = cos(90°) = **0 (leading)**

$$X_C = 1/(\omega C) \quad | \quad I_m = V_m/X_C$$



Phasor: Pure C Circuit

5.8 Impedance Triangle

The **Impedance Triangle** is a right-angled triangle graphically representing the relationship between Resistance (R), Reactance (X), and total Impedance (Z) in an AC circuit.

Components:

- **Base (Resistance, R):** Horizontal side; dissipates energy.
- **Perpendicular (Reactance, X):** Vertical side; X_L (up) or X_C (down); stores/releases energy.
- **Hypotenuse (Impedance, Z):** Total opposition to current flow.

$$Z = \sqrt{R^2 + X^2} \quad [\text{Ohms}, \Omega]$$

$$\tan\phi = X/R \quad | \quad \cos\phi = R/Z \quad | \quad \sin\phi = X/Z$$

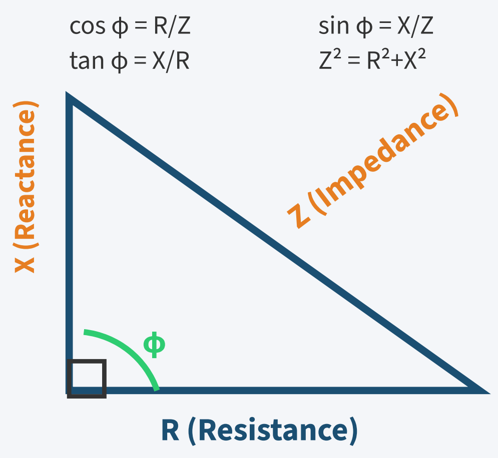


Fig: Impedance Triangle

5.9 Power in AC Circuits

ACTIVE POWER (P) – REAL POWER

Power actually consumed/utilized in the circuit. Measured in **Watts (W) or kW**.

$$P = VI\cos\phi = I^2R \; [\text{Watts}]$$

Also: $P = \sqrt{S^2 - Q^2}$

REACTIVE POWER (Q) – IMAGINARY POWER

Power that bounces between source and load. Measured in **VAR (Volt-Ampere Reactive)**.

$$Q = VI\sin\phi = I^2X \; [\text{VAR}]$$

Also: $Q = \sqrt{S^2 - P^2}$

APPARENT POWER (S) – TOTAL POWER

Product of RMS voltage and RMS current without phase angle. Measured in **VA or kVA**.

$$S = VI = \sqrt{P^2 + Q^2} \; [\text{VA}]$$

Also expressed as: $S = P + jQ$ (complex power)

Power Triangle

The power triangle is similar to the impedance triangle, with sides representing P, Q, and S.

- Base = Active Power (P) in kW
- Perpendicular = Reactive Power (Q) in kVAR
- Hypotenuse = Apparent Power (S) in kVA

```
kW + jkVAR = kVA
```

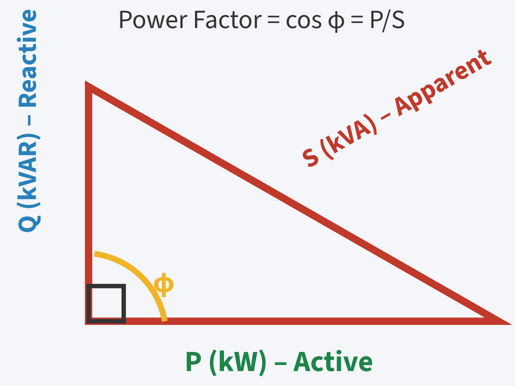


Fig: Power Triangle

5.10 Power Factor

Power factor is the ratio of active (real) power to apparent power, indicating how effectively incoming power is utilized.

```
Power Factor (PF) = cos ϕ = P/S = R/Z = Active Power / Apparent Power
```

- PF ranges from 0 to 1. **Unity PF (1)** → purely resistive; **Zero PF** → purely reactive.
- **Lagging PF:** Inductive load (current lags voltage) — motors, transformers.
- **Leading PF:** Capacitive load (current leads voltage) — capacitor banks.

6 Problem Solving – DC Circuits

DC Problem 1 – Ohm's Law Application

Easy

Problem: An electric heater draws 8A from a 230V supply. Find the resistance of the heater element and its power rating.

Given: $I = 8A$, $V = 230V$

Step 1 – Find Resistance:

By Ohm's Law: $V = IR \rightarrow R = V/I = 230/8$

$$R = 28.75\ \Omega$$

Step 2 – Find Power Rating:

$P = VI = 230 \times 8$

$$P = 1840\ W = 1.84\ kW$$

DC Problem 2 – KCL & KVL (Find Voltage across 4Ω Resistor)

Medium

Problem: In a circuit with a 6V battery, 8Ω and 4Ω resistors in series, find the voltage across the 4Ω resistor.

Given: $V = 6V$, $R_1 = 8\Omega$, $R_2 = 4\Omega$ (series)

Step 1 – Total Resistance:

$R_{total} = R_1 + R_2 = 8 + 4 = 12\Omega$

Step 2 – Total Current (KVL):

$V = I \times R_{total} \rightarrow I = 6/12 = 0.5A$

Step 3 – Voltage across 4Ω:

$V_2 = I \times R_2 = 0.5 \times 4$

$$V_{4\Omega} = 2\ V$$

DC Problem 3 – Parallel Circuit Analysis

Medium

Problem: Three resistors $R_1=6\Omega$, $R_2=3\Omega$, $R_3=2\Omega$ are connected in parallel across a 12V supply. Find (a) equivalent resistance, (b) total current, (c) current in each branch.

Step 1: $1/R_{eq} = 1/6 + 1/3 + 1/2 = 1/6 + 2/6 + 3/6 = 6/6 = 1 \rightarrow$ $\mathbf{R_{eq} = 1\Omega}$

Step 2: $I_{total} = V/R_{eq} = 12/1 =$ **12A**

Step 3: $I_1 = 12/6 = 2A$; $I_2 = 12/3 = 4A$; $I_3 = 12/2 = 6A$

$$I_1 + I_2 + I_3 = 2 + 4 + 6 = 12A \checkmark \text{ (KCL verified)}$$

DC Problem 4 – Superposition Theorem

Medium–Exam Level

Problem: Find current through 4Ω resistor in circuit with 20V voltage source and 5A current source. (6Ω in series with 20V; 4Ω in middle; 8Ω in series with 5A)

Step 1 (20V alone, 5A → open circuit):

Circuit: 20V — 6Ω — 4Ω in series

$R_{total} = 6 + 4 = 10\Omega$; $I_1 = 20/10 =$ **2A**

Step 2 (5A alone, 20V → short circuit):

6Ω and 4Ω in parallel: $R_{parallel} = (6\times4)/(6+4) = 2.4\Omega$

$V_{node} = 5 \times 2.4 = 12V$; $I_2 = 12/4 =$ **3A**

Step 3 – Superposition:

$I_{4\Omega} = I_1 + I_2 = 2 + 3$

$$I_{4\Omega} = 4 \text{ Amperes (Net current)}$$

DC Problem 5 – Series-Parallel Combination

Exam Level

Problem: Find the equivalent resistance between terminals A and B: $R_1=4\Omega$ series with parallel combination of $R_2=6\Omega$ and $R_3=3\Omega$.

Step 1 – Parallel part: $R_{23} = (6\times3)/(6+3) = 18/9 =$ **2Ω**

Step 2 – Series with R_1: $R_{AB} = R_1 + R_{23} = 4 + 2$

$$R_{AB} = 6\ \Omega$$

6 Problem Solving – AC Circuits

AC Problem 1 – RMS and Average Values

Easy

Problem: An AC voltage has a peak value of 325.27V and frequency 50 Hz. Find (a) RMS value, (b) Average value, (c) Time period.

(a) RMS: $V_{rms} = V_m/\sqrt{2} = 325.27/1.414 =$ **230 V**

(b) Average: $V_{avg} = 0.637 \times V_m = 0.637 \times 325.27 =$ **207.2 V**

(c) Time Period: $T = 1/f = 1/50 =$ **0.02 s = 20 ms**

AC Problem 2 – Form Factor & Peak Factor

Easy

Problem: For a sinusoidal voltage of 100V peak, find form factor and peak factor.

RMS: $V_{rms} = 100/\sqrt{2} = 70.7V$

Average: $V_{avg} = 2\times100/\pi = 63.7V$

Form Factor: $K_f = V_{rms}/V_{avg} = 70.7/63.7 =$ **1.11**

Peak Factor: $K_p = V_m/V_{rms} = 100/70.7 =$ **1.414 = $\sqrt{2}$**

AC Problem 3 – Inductive Reactance and Current

Medium

Problem: An inductor of 0.1H is connected to 230V, 50Hz AC supply. Find (a) inductive reactance, (b) rms current, (c) instantaneous current equation.

(a) X_L: $X_L = 2\pi fL = 2\pi \times 50 \times 0.1 =$ **31.42 Ω**

(b) I_{rms}: $I_{rms} = V_{rms}/X_L = 230/31.42 =$ **7.32 A**

(c) I_m: $I_m = \sqrt{2} \times 7.32 = 10.35A \rightarrow$ **$i(t) = 10.35 \sin(314t - \pi/2)$ A**

AC Problem 4 – Power Factor and Power Calculation

Medium–Exam Level

Problem: A series RL circuit has $R=10\Omega$, $L=0.05H$ connected to 230V, 50Hz supply. Find Z, I, PF, P, Q, and S.

X_L: $X_L = 2\pi \times 50 \times 0.05 = 15.71\Omega$

Z: $Z = \sqrt{(R^2 + X_L^2)} = \sqrt{(100 + 246.8)} = \sqrt{346.8} =$ **$18.62\ \Omega$**

I_{rms}: $I = V/Z = 230/18.62 =$ **12.35 A**

PF: $\cos\phi = R/Z = 10/18.62 =$ **0.537 (lagging)**

P: $P = I^2R = (12.35)^2 \times 10 =$ **1525.2 W ≈ 1.525 kW**

Q: $Q = I^2X_L = (12.35)^2 \times 15.71 =$ **2394.8 VAR ≈ 2.39 kVAR**

S: $S = VI = 230 \times 12.35 =$ **2840.5 VA ≈ 2.84 kVA**

AC Problem 5 – Capacitive Reactance

Exam Level

Problem: A capacitor of 100μF is connected to 230V, 50Hz AC supply. Find X_C, I_{rms}, and write the equation of current.

X_C: $X_C = 1/(2\pi fC) = 1/(2\pi\times50\times100\times10^{-6}) = 1/(0.03142) =$ **$31.83\ \Omega$**

I_{rms}: $I_{rms} = V/X_C = 230/31.83 =$ **7.23 A**

I_m: $I_m = \sqrt{2} \times 7.23 = 10.22A$

```
i(t) = 10.22 sin(314t + π/2) A  [Current leads voltage by 90°]
```

7 2-Mark Questions with Answers

Q1. Define Ohm's Law. State its mathematical form.

Ohm's Law states that the current through a conductor is directly proportional to the voltage across it, provided temperature remains constant. Mathematically: $V = IR$, where V is voltage (Volts), I is current (Amperes), and R is resistance (Ohms). The ratio $V/I = R$ remains constant for ohmic conductors.

Q2. What is KCL? State with sign convention.

Kirchhoff's Current Law (KCL): The algebraic sum of all currents at any node/junction is zero: $\Sigma I = 0$. Sign convention: Currents entering a node are positive (+); currents leaving are negative (−). This is based on the law of conservation of charge — charge cannot accumulate at a node.

Q3. What is KVL? Give the mathematical expression.

Kirchhoff's Voltage Law (KVL): In any closed loop, the algebraic sum of all EMFs equals the algebraic sum of all voltage drops: $\Sigma E = \Sigma IR$. Based on conservation of energy — no energy is lost or gained going around a closed loop. Positive for EMF rise, negative for voltage drops in direction of current.

Q4. Define RMS value of an AC quantity.

Root Mean Square (RMS) value of an AC quantity is the equivalent DC value that produces the same amount of heat (power) in a resistor over one complete cycle. For sinusoidal waveform: $V_{rms} = V_m/\sqrt{2} = 0.707\ V_m$. The standard household 230V AC refers to RMS value.

Q5. Define Average value of AC. Why is it taken over half cycle?

Average value is the mean of all instantaneous values over a period. For a complete AC cycle, the average is ZERO because positive and negative half cycles cancel. Hence average is taken over half cycle: $V_{av} = 2V_m/\pi = 0.637\ V_m$. This is useful for rectifier circuit analysis.

Q6. What is Form Factor and Peak Factor?

Form Factor (K_f) = V_{rms}/V_{avg} = 1.11 for sinusoidal wave. It indicates waveform shape (sharper peaks = higher form factor). Peak Factor (K_p) = $V_{peak}/V_{rms} = \sqrt{2} = 1.414$ for sinusoidal wave. It indicates the ratio of peak to effective value, important for insulation design.

Q7. What is Power Factor? What are its significance?

Power Factor = cos ϕ = Active Power/Apparent Power = R/Z. It ranges from 0 to 1. Unity PF means all power is consumed (pure resistive load). Low PF means poor utilization of electrical energy, higher current for same power, increased losses in cables, and penalization by electricity boards. Capacitor banks are used to improve (correct) lagging PF.

Q8. Differentiate Active, Reactive, and Apparent Power.

Active Power (P): Actual power consumed = VI cos ϕ (watts). Reactive Power (Q): Power stored and returned = VI sin ϕ (VAR) — does no useful work. Apparent Power (S): Total power = VI (VA) = $\sqrt{(P^2 + Q^2)}$. These form the power triangle where S is hypotenuse, P is base, and Q is the vertical side.

Q9. State the limitations of Ohm's Law.

Ohm's Law is not applicable to: (i) semiconductors and diodes, (ii) thermistors and vacuum tubes, (iii) electrolytic conductors, (iv) conditions where temperature/pressure are not constant, (v) non-linear devices where V-I graph is not a straight line. It applies only to good metallic conductors under constant physical conditions.

Q10. Define Frequency, Time Period, and Angular Frequency of AC.

Frequency (f): Number of complete cycles per second, measured in Hertz (Hz). f = 1/T. Time Period (T): Time taken to complete one full cycle, T = 1/f (seconds). Angular Frequency (ω): Rate of change of the phase angle, $\omega = 2\pi f$ radians/second. In India, standard AC frequency is 50 Hz, so T = 20ms and ω = 314 rad/s.

Q11. What is a phasor? How does it represent AC quantities?

A phasor is a rotating line of definite length (= peak value) rotating anticlockwise at constant angular velocity ω. Its projection on vertical axis at any instant gives the instantaneous value of the AC quantity. Phasor diagrams simplify AC analysis by representing magnitude and phase simultaneously using vectors, avoiding complex trigonometric computations.

Q12. Define Inductive Reactance. On what factors does it depend?

Inductive Reactance (X_L) is the opposition offered by an inductor to alternating current. $X_L = \omega L = 2\pi fL$ (Ohms). It depends on: (i) Frequency f — increases with f; (ii) Inductance L — directly proportional. At DC (f=0): X_L=0 (short circuit). At high frequency: X_L is large (open circuit). Voltage leads current by 90° in pure inductive circuit.

Q13. Define Capacitive Reactance. On what factors does it depend?

Capacitive Reactance (X_C) is the opposition offered by a capacitor to AC. $X_C = 1/(\omega C) = 1/(2\pi fC)$ (Ohms). It depends on: (i) Frequency f — decreases with increasing f; (ii) Capacitance C — inversely proportional. At DC (f=0): $X_C=\infty$ (open circuit). At high frequency: $X_C \to 0$. Current leads voltage by 90° in pure capacitive circuit.

Q14. What is Impedance? Write its formula for RL and RC series circuits.

Impedance (Z) is the total opposition offered by an AC circuit to the flow of alternating current, combining both resistance and reactance. For RL series: $Z = \sqrt{(R^2 + X_L^2)}$, $\phi = \tan^{-1}(X_L/R)$ lagging. For RC series: Z = $\sqrt{}$(R² +

X_C^2), $\phi = \tan^{-1}(X_C/R)$ leading. Unit: Ohms (Ω).

Q15. State Superposition Theorem with its conditions of applicability.

Superposition Theorem: In a linear network having multiple independent sources, the response at any element equals the algebraic sum of responses due to each source acting individually. Conditions: (i) Applicable only to LINEAR circuits, (ii) Valid for current and voltage only (NOT power), (iii) Voltage sources are short-circuited and current sources are open-circuited when not in use, (iv) Dependent sources must NOT be deactivated.

Q16. What is the phase difference between voltage and current in a pure resistor? Pure inductor? Pure capacitor?

Pure Resistor: Phase difference = 0° (in phase). PF = cos 0° = 1. Pure Inductor: Phase difference = 90°, voltage leads current. PF = cos 90° = 0 (lagging). Pure Capacitor: Phase difference = 90°, current leads voltage. PF = cos 90° = 0 (leading). Memory trick: "ELI the ICE man" — E leads I in L (Inductor); I leads E in C (Capacitor).

Q17. Define Amplitude and Instantaneous value of an AC waveform.

Amplitude (V_m): The maximum value attained by an alternating quantity during positive or negative half cycle. Also called peak value. Instantaneous Value (v): The value of an AC quantity at a specific instant of time: $v = V_m \sin(\omega t)$. It continuously changes with time. At t=0, v=0; at $\omega t=\pi/2$, $v=V_m$ (maximum).

Q18. What is the function of a Resistor, Inductor, and Capacitor in a circuit?

Resistor (R): Limits current, converts electrical energy to heat (Joule heating), used for current control and voltage division. Inductor (L): Stores energy in magnetic field, opposes change in current, used in filters and transformers. Capacitor (C): Stores energy in electric field, blocks DC, passes AC, used for power factor correction, timing circuits, and energy storage.

Q19. What is Peak-to-Peak voltage? Relate it to amplitude.

Peak-to-Peak voltage (V_{pp}) is the total voltage swing from the maximum positive peak ($+V_m$) to the maximum negative peak ($-V_m$) of an AC waveform. $V_{pp} = 2 \times V_m = 2 \times$ amplitude. For a 230V RMS supply: $V_m = 325.27V$, so $V_{pp} = 650.54V$. Important in oscilloscope measurements and amplifier design.

Q20. Define equivalent resistance of series and parallel circuits.

Series: $R_{eq} = R_1 + R_2 + ... + R_n$ (always greater than the largest individual R). Parallel: $1/R_{eq} = 1/R_1 + 1/R_2 + ... + 1/R_n$ (always less than the smallest individual R). For two parallel resistors: $R_{eq} = R_1R_2/(R_1+R_2)$ (product over sum). These formulas are derived from KVL (series) and KCL (parallel) applications.

8 5-Mark Questions with Answers

Q1. Explain electrical circuit elements R, L, C with diagrams and compare their properties. [5M]

Resistor (R): A passive element that converts electrical energy to heat. Symbol: zig-zag (—/\/\/—). Governed by Ohm's Law: $V = IR$. Energy dissipation $P = I^2R$ (watts). Offers constant opposition regardless of frequency. No energy storage.

Inductor (L): A coil that stores energy in its magnetic field. Symbol: loops (—⌇⌇⌇⌇—). Governed by: $V = L(di/dt)$. $X_L = \omega L$ increases with frequency. At DC: acts as short circuit; at high f: acts as open. Voltage leads current by 90°.

Capacitor (C): Two plates separated by dielectric, stores charge/energy in electric field. Symbol: —| |—. Governed by: $I = C(dv/dt)$. $X_C = 1/\omega C$ decreases with frequency. At DC: open circuit; at high f: short circuit. Current leads voltage by 90°.

Key Comparison: R dissipates energy; L and C store and return energy. All three are passive linear elements (for ideal components). Impedance: $Z_R=R$, $Z_L=j\omega L$, $Z_C=1/j\omega C$.

Q2. State and explain Kirchhoff's Laws with an example for each. [5M]

KCL (Kirchhoff's Current Law): At any node, $\Sigma I = 0$ (sum of currents entering = sum leaving). Based on conservation of charge. Example: At node A, if $I_1=3A$ enters, $I_2=2A$ enters, then $I_3 = I_1+I_2 = 5A$ must leave. Equation: $I_1+I_2-I_3=0$.

KVL (Kirchhoff's Voltage Law): In any closed loop, $\Sigma E = \Sigma IR$. Based on conservation of energy — no potential difference around a closed loop. Sign Convention: EMF rise = +E; voltage drop in direction of $I = -IR$.

KVL Example: For a loop with $E_1=10V$, $E_2=4V$, $R_1=2\Omega$, $R_2=3\Omega$, all in series: KVL gives $E_1-E_2 = I(R_1+R_2) \rightarrow 10-4 = I\times5 \rightarrow I = 6/5 = 1.2A$. Voltage drop across $R_1 = 1.2\times2 = 2.4V$; across $R_2 = 1.2\times3 = 3.6V$. Check: $2.4+3.6 = 6V = E_1-E_2$ ✓

These laws are fundamental tools for mesh analysis and nodal analysis of complex circuits.

Q3. Derive and explain the RMS value of an alternating current. [5M]

Definition: RMS (Root Mean Square) value is the DC equivalent value producing same heating effect in a resistor as the AC waveform.

Derivation: Consider $i = I_m \sin(\omega t)$. The mean square value over one cycle:

$I^2_{rms} = (1/T) \int_0^T i^2\, dt = (1/T) \int_0^T I^2_m \sin^2(\omega t)\, dt$

Using $\sin^2(\theta) = (1-\cos2\theta)/2$:

$= (I^2_m/T) \int_0^T [(1-\cos2\omega t)/2]\, dt = (I^2_m/2T)\, [t - \sin2\omega t/2\omega]_0^T$

Substituting limits ($\sin2\omega T = \sin4\pi = 0$):

$I^2_{rms} = I^2_m/2 \rightarrow I_{rms} = I_m/\sqrt{2} = \mathbf{0.707\, I_m}$

Physical significance: Standard electricity supply is specified in RMS — 230V AC means V_{rms}=230V, so V_m=325.27V. RMS is used for power calculations: $P = I^2_{rms} \times R$.

Q4. Explain phasor diagrams for pure R, L, and C circuits with voltage-current waveforms. [5M]

Pure R: $v = V_m\sin(\omega t)$, $i = I_m\sin(\omega t)$. Both in phase (ϕ=0°). PF=1. Power P=VI. Phasor: V and I point in same direction. Waveforms overlap perfectly.

Pure L: $v = V_m\sin(\omega t)$, $i = I_m\sin(\omega t-\pi/2)$. Voltage LEADS current by 90°. $X_L=\omega L$. PF=0 (lagging). Power consumed = 0. Phasor: V is horizontal; I points downward at 90°. ELI: In L, E leads I.

Pure C: $v = V_m\sin(\omega t)$, $i = I_m\sin(\omega t+\pi/2)$. Current LEADS voltage by 90°. $X_C=1/\omega C$. PF=0 (leading). Power consumed = 0. Phasor: V is horizontal; I points upward at 90°. ICE: In C, I leads E.

Memory: **"ELI the ICE man"** — E leads I in Inductors (L); I leads E in Capacitors (C).

Q5. Explain Superposition Theorem with step-by-step procedure and an example. [5M]

Statement: In a linear bilateral network having multiple independent sources, the current/voltage through any element = algebraic sum of currents/voltages due to each source acting independently.

Procedure: (1) Take one independent source; short all other voltage sources (R_{int}≈0) and open all current sources ($R_{int}=\infty$). (2) Find the response. (3) Repeat for each source. (4) Add all responses algebraically.

Example: Circuit: 20V source with 6Ω and 4Ω. Due to 20V: I_1(4Ω) = 20/(6+4) = 2A. Due to 5A current source (20V shorted): parallel 6Ω||4Ω gives node voltage = 5×(6||4) = 5×2.4 = 12V. I_2(4Ω) = 12/4 = 3A. Total I = 2+3 = 5A.

Limitations: Not valid for power (non-linear calculation); not valid for non-linear circuits; dependent sources must remain active.

Q6. Draw and explain the impedance triangle. Derive expressions for Z, power factor, and phase angle. [5M]

The impedance triangle is a right-angled triangle for AC circuits. Base = R (resistance, Ω); Perpendicular = X (reactance, Ω); Hypotenuse = Z (total impedance, Ω).

For RL series: $Z = \sqrt{(R^2 + X_L^2)}$. Phase angle: $\phi = \tan^{-1}(X_L/R)$ — lagging (voltage leads current).

For RC series: $Z = \sqrt{(R^2 + X_C^2)}$. Phase angle: $\phi = \tan^{-1}(X_C/R)$ — leading (current leads voltage).

Power Factor: $\cos\phi = R/Z$ (from triangle). Also PF = Active Power/Apparent Power = P/S.

Relations: $R = Z\cos\phi$; $X = Z\sin\phi$; $\tan\phi = X/R$. The impedance triangle is geometrically similar to the voltage triangle (V_R, V_X, V) in series AC circuits.

Example: R=6Ω, X_L=8Ω: $Z=\sqrt{(36+64)}$=10Ω; $\cos\phi$ = 6/10 = 0.6; ϕ = 53.13°.

Q7. Explain Active, Reactive, and Apparent Power with power triangle. [5M]

Active Power (P): $P = VI\cos\phi = I^2R$ (Watts/kW). Actual power consumed and converted to useful work (heat, mechanical, light). Measured by wattmeter. This is the power we pay for in electricity bills.

Reactive Power (Q): $Q = VI \sin \phi = I^2X$ (VAR/kVAR). Power that alternates between source and load without doing useful work. Caused by inductors and capacitors. Essential for maintaining magnetic/electric fields in motors, transformers.

Apparent Power (S): $S = VI = I^2Z$ (VA/kVA). Total power supplied by source. Vector sum: $S = \sqrt{P^2+Q^2}$. Power triangle: S is hypotenuse, P is base, Q is perpendicular. Power factor = $\cos \phi = P/S$ = kW/kVA. High PF means less reactive component — more efficient system.

Q8. Derive the average value of an alternating sinusoidal current. [5M]

Definition: Average value is the mean of all instantaneous values over a half period (full cycle average = 0).

Derivation: Let $i = I_m \sin(\omega t)$. Over half cycle (0 to T/2):

$I_{avg} = (1/(T/2)) \int_0^{T/2} I_m \sin(\omega t)\, dt = (2/T) \int_0^{T/2} I_m \sin(\omega t)\, dt$

$= (2I_m/T) \times [-\cos(\omega t)/\omega]_0^{T/2}$

$= (2I_m/T\omega) \times [-\cos(\omega T/2) + \cos(0)]$

Since $\omega = 2\pi/T \rightarrow \omega T/2 = \pi$, $\cos(\pi) = -1$, $\cos(0) = 1$:

$I_{avg} = (2I_m/T\omega) \times [1+1] = (2I_m \times 2)/(T \times 2\pi/T) = 4I_m/(2\pi)$

$\mathbf{I_{avg} = 2I_m/\pi = 0.637\, I_m}$

This value is used in rectifier analysis and in the calculation of Form Factor.

Q9. Explain series and parallel circuits with derivation of equivalent resistance. [5M]

Series Circuit: Resistors connected end-to-end; same current I flows through all. Applying KVL: $V = V_1+V_2+V_3 = IR_1+IR_2+IR_3 = I(R_1+R_2+R_3)$. Therefore $\mathbf{R_{eq} = R_1+R_2+...+R_n}$. Total resistance increases, always greater than any individual resistor. Voltage divider principle: $V_1/V_2 = R_1/R_2$.

Parallel Circuit: All resistors connected between same two nodes; same voltage V across all. Applying KCL: $I = I_1+I_2+I_3 = V/R_1+V/R_2+V/R_3 = V(1/R_1+1/R_2+1/R_3)$. Therefore $\mathbf{1/R_{eq} = 1/R_1+1/R_2+...+1/R_n}$. Equivalent R is always less than smallest branch R. Current divider: $I_1/I_2 = R_2/R_1$.

Applications: Series — voltage divider, fuse circuits. Parallel — domestic wiring, household appliances (independent switching).

Q10. What is power factor? Explain its significance and types. [5M]

Definition: Power Factor = $\cos \phi = P/S = R/Z$ = Active Power/Apparent Power. It measures how effectively electrical power is being utilized.

Types: (i) Unity PF ($\cos \phi=1$): Pure resistive load; all power consumed. (ii) Lagging PF ($0 < \cos \phi < 1$ lagging): Inductive load like motors, transformers; current lags voltage. (iii) Leading PF ($0 < \cos \phi < 1$ leading): Capacitive load; current leads voltage.

Significance: Low PF increases current for same power delivery → higher cable losses (I^2R), larger conductor sizes, larger KVA rating of generators/transformers, increased electricity bills (penalty for industrial consumers). Electricity boards charge penalty for PF below 0.8 lagging.

Improvement: PF can be improved by connecting capacitor banks in parallel with inductive loads. This reduces reactive power demand from the source and brings PF closer to unity.

9 Previous Year Questions (PYQs)

Q1. Explain electrical circuit elements R, L, C with neat diagrams. Discuss their energy handling behaviour.

Easy | Asked 5 times | ☆ Most Repeated

Q2. State and explain Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) with suitable examples.

Medium | Asked 6 times | ☆ Very High Priority

Q3. Derive the expression for RMS value of an alternating sinusoidal current with necessary waveforms. [5M]

Medium | Asked 5 times

Q4. Derive and explain the average value (mean value) of an alternating current. [5M]

Medium | Asked 4 times

Q5. Explain peak factor and form factor of a periodic waveform with expressions.

Easy | Asked 4 times

Q6. Derive the expressions for voltage and current relationships in pure R, L, and C circuits. Draw phasor diagrams for each.

Hard | Asked 6 times | ☆ Highest Priority

Q7. Draw and explain the impedance triangle. Derive expressions for impedance, phase angle, and power factor.

Medium | Asked 4 times

Q8. Explain Active Power, Reactive Power, and Apparent Power with the power triangle. [5M]

Medium | Asked 5 times

Q9. State and verify Superposition Theorem using a suitable DC circuit example. [10M]

Hard | Asked 4 times

Q10. An AC circuit has R=6Ω and X_L=8Ω connected in series to 100V AC supply. Find Z, I, PF, active power, and reactive power.

Hard | Asked 5 times | ⭐ Numerical — Very Important

Answer: Z=10Ω, I=10A, PF=0.6 lagging, P=600W, Q=800VAR, S=1000VA

10 Exam Tips & Strategy

⏱ Time Management

- ▶ Read all questions first (5 min). Identify easy, medium, hard.
- ▶ Allocate: 2M questions → 3 min each; 5M → 8 min each; 10M → 18 min.
- ▶ Attempt all 2M questions first — guaranteed marks.
- ▶ Reserve 10 min for revision at the end.

📝 Numerical Strategy

- ▶ Always write "Given:" and "To Find:" clearly.
- ▶ State the formula before substituting values.
- ▶ Show every step — partial marks are awarded.
- ▶ Box your final answer. Write correct units.
- ▶ Verify using check equations (KCL, KVL).

⚠ Common Mistakes to Avoid

- ▶ Confusing V_m with V_{rms} in problems.
- ▶ Forgetting phase angle sign (+ for leading, − for lagging).
- ▶ Using Superposition Theorem for power calculations (invalid).
- ▶ Not converting units (μF to F, mH to H).
- ▶ Forgetting to draw phasor diagram in AC questions.

✅ Presentation Tips

- ▶ Draw neat, labeled circuit diagrams for every Q.
- ▶ Use phasor diagrams in AC circuit questions.
- ▶ Write formulas in highlighted boxes where possible.
- ▶ Use arrows for current direction in diagrams.
- ▶ Underline key terms and final answers.

11 Quick Revision Sheet – All Formulas

DC Circuit Laws

- $V = IR$ (Ohm's Law)
- $P = VI = I^2R = V^2/R$
- $\Sigma I = 0$ at any node (KCL)
- $\Sigma E = \Sigma IR$ in any loop (KVL)
- $R_{series} = R_1+R_2+R_3$
- $1/R_{parallel} = 1/R_1+1/R_2+1/R_3$

AC Waveform Parameters

- $v = V_m \sin(\omega t)$; $i = I_m \sin(\omega t)$
- $\omega = 2\pi f$; $f = 1/T$
- $V_{rms} = V_m/\sqrt{2} = 0.707V_m$
- $V_{avg} = 2V_m/\pi = 0.637V_m$
- Form Factor = Vrms/Vavg = 1.11
- Peak Factor = V_m/Vrms = $\sqrt{2}$ = 1.414

Reactance & Impedance

- $X_L = \omega L = 2\pi fL$ (Ω)
- $X_C = 1/\omega C = 1/2\pi fC$ (Ω)
- $Z = \sqrt{R^2 + X^2}$
- $\tan \phi = X/R$
- $\cos \phi = R/Z$ (Power Factor)
- $V = I \cdot Z$ (AC Ohm's Law)

Power Formulas

- $P = VI \cos \phi$ (Active, Watts)
- $Q = VI \sin \phi$ (Reactive, VAR)
- $S = VI$ (Apparent, VA)
- $S = \sqrt{P^2 + Q^2}$
- $PF = \cos \phi = P/S = R/Z$
- $P = I^2R$; $Q = I^2X$; $S = I^2Z$

12 Additional Topics – Resonance & Practical Applications

Basic Idea of Resonance

Resonance in an AC circuit occurs when the inductive reactance equals the capacitive reactance: $\mathbf{X_L = X_C}$. At resonance, the net reactance is zero, impedance is purely resistive and minimum ($Z = R$), and current is maximum.

$$\text{Resonant Frequency: } f_0 = 1/(2\pi\sqrt{LC}) \text{ Hz}$$

- At resonance: $X_L = X_C \rightarrow \omega_0 L = 1/\omega_0 C$
- Current is maximum; power factor = 1 (unity)
- Applications: Radio tuning circuits, filters, oscillators

Practical Applications

Concept	Practical Application

Resistor (R)	Heaters, lighting, fuses, current limiters, voltage dividers
Inductor (L)	Motors, transformers, choke coils, SMPS inductors
Capacitor (C)	Power factor correction, bypass filters, timing circuits, energy storage
KCL/KVL	Circuit simulation (SPICE), fault analysis, power system protection
Superposition	Linear amplifier analysis, multi-source power systems
AC Power	Electricity metering (kWh), power factor penalty billing, FACTS devices
Resonance	AM/FM radio tuning, bandpass filters, MRI machines

💡 **Power Triangle Insight:** For an industrial consumer drawing 100 kVA at PF=0.8 lagging: P = 80 kW (paid power), Q = 60 kVAR (reactive burden), S = 100 kVA (total apparent). If PF is improved to unity with capacitors, only 80 kW = 80 kVA is drawn — 20% reduction in current for same work output!

13 Final Practice Section

Short Answer Practice

1. Define one coulomb of charge. How many electrons make one coulomb?
2. What is the difference between EMF and terminal voltage?
3. An 8Ω resistor is connected to 24V DC. Find (a) current, (b) power consumed.
4. State the condition for maximum current in an AC series circuit.
5. What will be the reactance of a 0.2H inductor at 50 Hz? At 100 Hz?
6. Why does X_C decrease as frequency increases? Explain physically.
7. A capacitor allows AC but blocks DC — explain with reactance formula.
8. If V_{rms} = 220V, what is the peak voltage? Peak-to-peak voltage?
9. Find the equivalent resistance of 6Ω and 3Ω in parallel.
10. What is the significance of unity power factor in AC systems?

Long Answer Practice

1. A series circuit has R=10Ω, L=31.8mH, C=318μF connected to 230V, 50Hz. Find Z, I, PF, P, Q, S, and phase angle. Draw phasor diagram.
2. Using KVL, find the current in each branch of a bridge circuit with E=10V, R_1=2Ω, R_2=3Ω, R_3=4Ω, R_4=5Ω.
3. Find the current through the 5Ω resistor using Superposition Theorem. Given: 10V and 5A source, 5Ω, 10Ω resistors in the network.
4. Derive the expression for the power consumed in a series RL circuit and prove that reactive power is zero in a pure inductor.
5. Explain how power factor can be improved using capacitor banks. Derive the expression for required capacitance to improve PF from cos ϕ_1 to cos ϕ_2.

PYQ with Hints

PYQ: Find current & power in RL series circuit [JNTUK 2022] Exam Level

Given: R=6Ω, X_L=8Ω, V=100V (AC, 50Hz)

Z: $\sqrt{(6^2+8^2)} = \sqrt{100}$ = **10 Ω**

I: V/Z = 100/10 = **10 A**

PF: cos ϕ = R/Z = 6/10 = **0.6 lagging**; ϕ = 53.13°

P: VI cos ϕ = 100×10×0.6 = **600 W**

Q: VI sin ϕ = 100×10×0.8 = **800 VAR**

S: VI = 100×10 = **1000 VA**

14 Conclusion – Unit Summary

Unit 1 of BEEE (JNTUK R23) covers the foundational principles of both DC and AC electrical circuits. The key takeaways are:

- ▶ **R, L, C Elements:** Three passive elements with distinct energy behaviours — R dissipates, L stores magnetically, C stores electrically. Their combination determines AC circuit behaviour.
- ▶ **DC Laws:** Ohm's Law (V=IR) and Kirchhoff's Laws (KCL: ΣI=0; KVL: ΣE=ΣIR) form the backbone of circuit analysis, enabling systematic solution of any linear network.
- ▶ **Circuit Types:** Series (same I), Parallel (same V), and Series-Parallel combinations are analysed using equivalent resistance concepts.
- ▶ **Superposition Theorem:** Powerful tool for multi-source linear circuits, applicable only to voltage/current (not power) in linear bilateral circuits.
- ▶ **AC Fundamentals:** Sinusoidal waveforms characterised by amplitude, frequency, phase, and angular frequency. $V_{rms}=0.707V_m$; $V_{avg}=0.637V_m$.
- ▶ **Phasor Analysis:** V leads I by 90° in inductors (ELI); I leads V by 90° in capacitors (ICE). Resistors have unity PF with V and I in phase.
- ▶ **Power Triangle:** P (Active, kW) + Q (Reactive, kVAR) = S (Apparent, kVA). PF = cos ϕ = P/S. Higher PF means more efficient power utilization.

🎯 **EXAM FOCUS SUMMARY**

Most asked topics: Phasor diagrams (R,L,C) | KCL & KVL with numericals | RMS derivation | Power triangle | Superposition Theorem | Form & Peak factors. Prepare at least 2 numerical examples for each topic. Draw diagrams for every theory question. Remember: **ELI the ICE man** for phase relationships!

All the Best for Your Examinations!

BEEE Unit 1 | JNTUK R23 | DC & AC Circuits

Study Smart · Practice Problems · Revise Formulas · Score High